Brain-Inspired Optimization Regime for Continual Artificial Intelligence

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Abstract

The abstract serves both as a general introduction to the topic and as a brief, non-technical summary of the main results and their implications. Authors are advised to check the author instructions for the journal they are submitting to for word limits and if structural elements like subheadings, citations, or equations are permitted.

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1 Introduction

Artificial intelligence is increasingly expected to operate in domains where knowledge must be updated continuously, yet most machine learning pipelines remain optimized under sequential, non-stationary updates. Consequently, adaptation to new tasks often degrades performance on earlier ones, a phenomenon known as catastrophic forgetting. In these settings, forgetting is not merely a technical limitation, it is a fundamental barrier to AI systems whose value depends on the reliable accumulation of knowledge.

A substantial body of work, including memory replay, parameter isolation and regularization, has sought to mitigate catastrophic forgetting. Although these approaches have yielded important progress, they often target specific failure modes rather than the broader requirements of durable adaptation. In practice, continual learning is shaped by at least three recurring challenges: i) New updates can interfere destructively with representations that support prior competence. ii) Retained knowledge can remain encoded in geometrically fragile parameter states that are easily destabilized by subsequent optimization. iii) In addition, many realistic adaptation settings, especially those involving constrained resources, limited system access or delayed and expensive feedback, do not permit efficient reliance on full backpropagated gradients. A durable continual learner must therefore do more than preserve benchmark performance. It must suppress interference, stabilize memory and remain adaptable when exact gradient information is costly or unavailable.

Biological nervous systems provide a useful functional reference for durable adaptation. Rather than preserving prior knowledge by freezing plasticity, the mammalian brain regulates

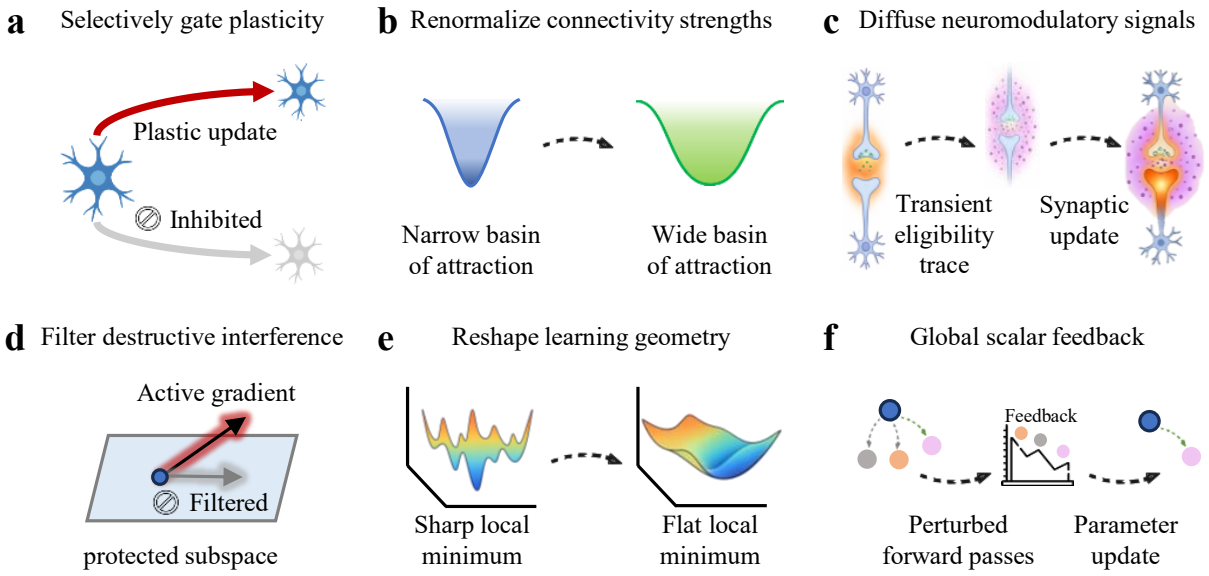

Fig. 1: Biological principles and computational analogues in INFTY. a–c, Biological motifs suggesting three functional constraints for continual adaptation: selective plasticity, stable memory and learning under limited feedback. d–f, Corresponding optimization mechanisms in INFTY: gradient filtering to suppress destructive interference, geometry reshaping toward flatter minima, and zeroth-order updates from perturbed forward passes with scalar feedback. Together, these motifs frame continual adaptation as the negotiation of multiple recurring constraints rather than a single failure mode.

where change occurs, consolidates experience into more robust and distributed states, and supports adaptation through local activity combined with modulatory feedback. These features indicate recurring requirements for continual AI: updates should be selective rather than globally disruptive, memory should be stabilized against subsequent perturbation, and adaptation should remain feasible when precise gradient signals are limited. From this perspective, biology provides a principled functional specification of the constraints that any durable learning system must negotiate under persistent change.

Here we propose INFTY, a biologically motivated optimization framework that organizes continual adaptation around three complementary modules corresponding to these requirements. To reduce destructive interference, INFTY selectively reshapes update directions. To improve retention, it steers optimization toward locally flatter solutions. When gradient access is limited, it supports adaptation through zeroth-order estimation based on stochastic perturbations and forward-pass feedback. Although these modules are independently deployable, they are not an arbitrary collection of techniques. Rather, they represent a structured decomposition of three recurrent constraints on continual adaptation: selective updating, stable retention and feasible learning under limited feedback.

Viewed in this way, continual learning is not governed by a single source of forgetting, but by multiple constraints on how new information can be incorporated without eroding prior competence. Across diverse continual learning settings, including resource-constrained and feedback-limited regimes, we evaluate these modules separately and show that each improves a distinct requirement of durable adaptation. More broadly, INFTY suggests that reliable continual AI depends not on a single universal remedy, but on adaptation strategies matched to the constraints that limit learning.

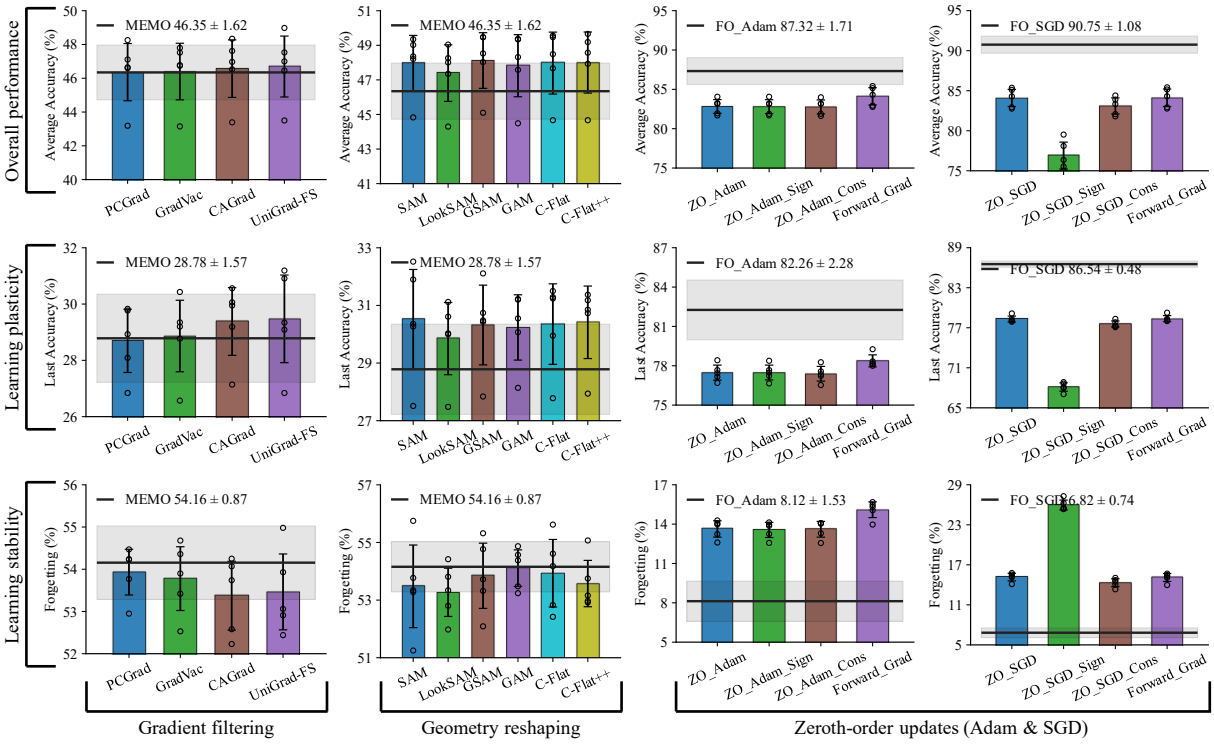

Fig. 2: Experimental results. The evaluation metrics include average accuracy (top), learning plasticity (middle), and stability (bottom). The tested methods include gradient filtering (left), geometry reshaping (center), and zeroth-order updates (right). All results are averaged over five runs with different random seeds and task orders. Error bars represent the standard error of the mean.

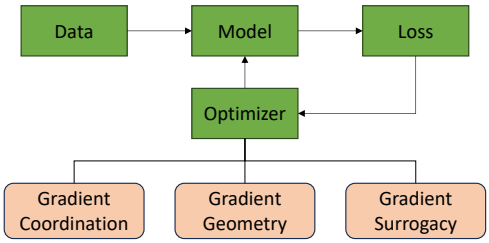

Fig. 3: Split of optimization

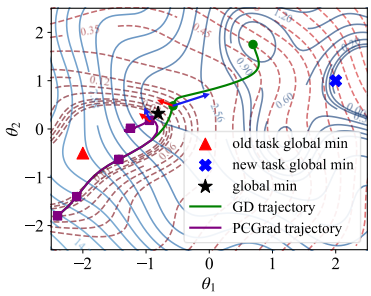

Fig. 4: Illustration of gradient conflicts. Gradient descent (GD) is dominated by the strongest gradient direction, whereas PCGrad [1] mitigates interference by removing conflicting components, enabling more effective global search.

2 Results

2.1 Gradient filtering suppresses destructive interference

In continual learning, updates for new tasks can interfere with parameter directions that support previously acquired competence, creating a direct trade-off between plasticity and retention. A related principle is observed in biological systems, where plasticity is selectively gated so that new learning does not globally disrupt existing representations. This functional analogy motivates gradient filtering methods, which reduce destructive interference by suppressing update components that are incompatible with earlier knowledge.

Figure 4 illustrates this intuition. When gradients associated with plasticity and retention conflict, standard gradient descent tends to follow the dominant direction, which can degrade the competing objective and bias optimization away from solutions that balance both requirements. By contrast, PCGrad [1] removes conflicting components and preserves only compatible directions, yielding an update that better reconciles adaptation with knowledge preservation. In this sense, gradient filtering acts as an optimization-level analogue of selective plasticity, allowing new learning to proceed while limiting disruption to established representations.

This perspective is reflected in a broader family of methods that coordinate competing update signals through projection, alignment or regularization. GradVac [2] encourages geometrically aligned updates for related tasks, whereas CAGrad [3] optimizes for worst-case local improvement with guarantees toward Pareto-stationary solutions. OGD [4] mitigates interference by projecting updates away from compressed historical gradient subspaces without requiring access to past data, and UniGrad-FS [5] further reduces conflicts by combining gradient similarity with flatness-aware optimization. As shown in Figure 2, applying these constraints to MEMO consistently reduces forgetting while preserving, and in some cases improving, average and last-task accuracy. In particular, PCGrad and GradVac maintain performance comparable to vanilla MEMO with better retention, whereas CAGrad and UniGrad-FS yield stronger final-task performance, suggesting that suppressing destructive interference can improve both stability and adaptation.

2.2 Geometry reshaping stabilizes memory retention

Even when previously acquired knowledge is retained, it may remain encoded in solutions that are geometrically fragile and therefore easily destabilized by later updates. Biological systems address a related problem not by freezing synapses, but by consolidating experience into more robust and distributed states through processes such as homeostatic regulation and memory consolidation. This functional analogy motivates flatness-oriented objectives in continual learning,

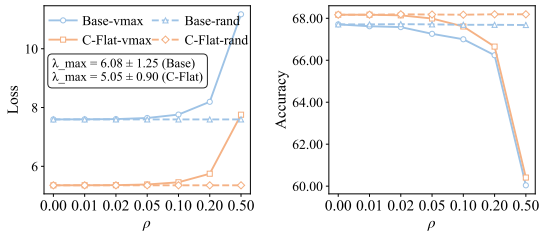

Fig. 5: Comparison of Hessian eigenvalues and loss/accuracy changes under ρ-scaled perturbations between C-Flat and the SGD baseline on MEMO with 10-split CIFAR-100. C-Flat converges to a noticeably flatter solution.

which aim to improve retention by steering optimization toward solutions that are less sensitive to subsequent parameter drift.

Figure 5 illustrates this effect for C-Flat on MEMO with 10-split CIFAR-100. Compared with vanilla SGD, C-Flat converges to a solution with a substantially smaller maximum Hessian eigenvalue, indicating a flatter local geometry. We further perturb the parameters along the dominant eigenvector and along a random direction with increasing ρ-scaled magnitudes, and measure the resulting changes in loss and accuracy. Within the prescribed perturbation radius, C-Flat shows reduced curvature in the worst-case direction and improved robustness to perturbations along random directions. Even at larger perturbation scales, where both methods become unstable along the dominant eigenvector, C-Flat continues to exhibit smaller fluctuations under random perturbations. These results indicate that geometry reshaping improves local stability and makes previously learned solutions more tolerant to the drift induced by continual adaptation.

A similar principle underlies a broader family of sharpness-aware methods that can be incorporated into continual learning. SAM [6] minimizes the worst-case local loss increase and already provides a strong flatness-aware baseline. Look-SAM [7] approximates perturbed gradients more efficiently by caching the sharpness-driven component, while GSAM [8] and GAM [9] introduce additional surrogate objectives to better control local geometry. C-Flat++ [10] further accelerates this strategy by avoiding unnecessary perturbation steps. As shown in Figure 2, these methods consistently outperform baselines that do

not explicitly control sharpness. Taken together, these results support a common view: by pushing optimization toward wider and more stable regions of the loss landscape, geometry reshaping improves tolerance to subsequent updates and thereby mitigates forgetting.

2.3 Zeroth-order updates enable adaptation under limited feedback

Unlike artificial neural networks, the biological brain does not have access to a precise, globally propagated error signal comparable to backpropagation. Instead, learning emerges from local Hebbian updates, stochastic synaptic transmission, and the interaction between eligibility traces and diffuse neuromodulatory signals. When an action yields a favorable outcome, neuromodulators such as dopamine are released and broadcast as a scalar reward signal, selectively reinforcing synapses that were recently active and thus tagged by eligibility traces. This corresponding to a learning paradigm: local variability in neural activity is explored first, and delayed global feedback subsequently determines which synaptic fluctuations are consolidated into long-term changes.

In deep learning, explicit gradients may be unavailable or deliberately disabled because of non-differentiable modules, frozen backbones for stability, or noisy task-level feedback. In such cases, the optimizer perturbs parameters stochastically and infers approximate descent directions solely from changes in performance. In our continual learning setting, zeroth-order gradient estimators probe the objective with randomized parameter perturbations and updates the weights according to the induced loss differences, thereby enabling gradient-free credit assignment across complex CL pipelines while respecting gradient constraints imposed to control forgetting. In Figure 2, we bypass explicit gradient computation by relying on local function evaluations of ZeroFlow [11] during the optimization of EASE. Among the zeroth-order optimizers, ZO-SGD [12] estimates descent directions by applying randomized parameter perturbations and measuring the induced loss differences. ZO-SGD-Sign [13] instead uses the sign of the gradient estimate, which allows for larger parameter steps but can also exacerbate convergence instability. ZO-SGD-Conserve [14] adopts a more conservative strategy by comparing the update direction at the current point with two candidate directions along and opposite to the estimated descent direction, at the cost of a more constrained search. Adam-based variants [15] additionally incorporate momentum, which typically leads to more stable convergence and better final performance compared with their SGD counterparts.

3 Discussion

Continual learning techniques enable AI models or agents to continuously adapt to open-world environments, serving as a critical step toward genuinely autonomous intelligence. While current research has proposed numerous continual learning frameworks (spanning CNN-based models, pretrained models, large language models and vision–language models, employing experience replay, regularization constraints, architectural expansion and prompting strategies, and covering task, domain, class and concept-incremental settings), these methods still mainly focus on predefined scenarios. When new settings arise, substantial effort is required to redesign data pipelines, model architectures, and loss functions in order to make them work effectively. However, the role of the optimizer in continual learning updates is often overlooked. In this paper, we draw inspiration from distinct biological mechanisms of learning in the brain to analyze and improve the optimization process in continual learning models. We focus on three families of mechanisms, namely synaptic regulation, memory consolidation, and reward-driven signaling, which capture the dynamic behavior of brain learning under intrinsic constraints, and we translate the resulting insights into optimization strategies for artificial systems.

Typical Continual Learning. First, from the perspective of gradient coordination, we abstract how synaptic changes arising from distinct neuronal populations that support new tasks are selectively gated. Local structural control plays the role of a filter that determines which parameters are allowed to update, thereby reducing interference at the architectural level, while neuromodulatory signals globally modulate the strength and direction of these synaptic changes,

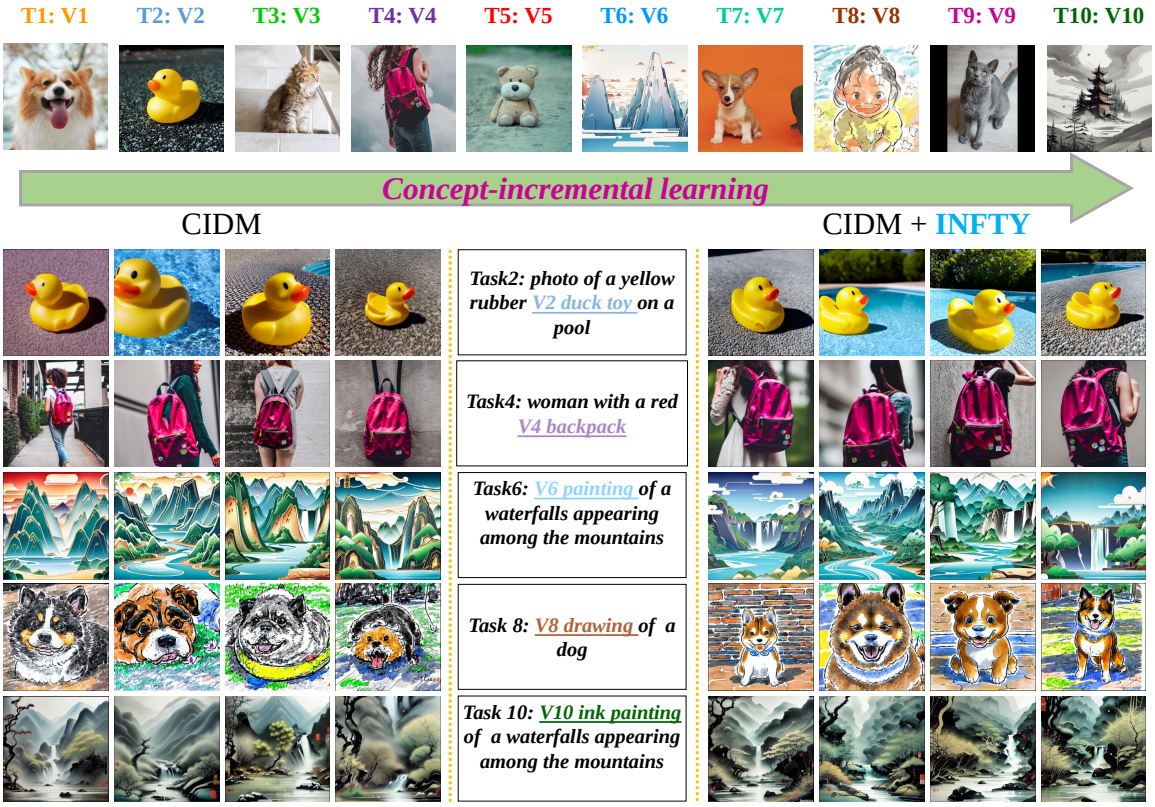

Fig. 6: Qualitative comparison of single-concept customizations generated by CIDM and INFTY.

Table 1: Performance on T1–T10 and average accuracy.

Method	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	Avg
DMNSP	**99.20**	96.10	**91.93**	87.05	87.00	86.10	84.17	83.05	81.58	79.94	87.61
+INFTY	**99.20**	**96.30**	91.80	**87.30**	**87.44**	**86.60**	**84.46**	**83.20**	**81.69**	**80.52**	**87.85**

further suppressing conflicts between overlapping functions. This view parallels multi-objective optimization methods that explicitly resolve gradient conflicts by adjusting gradient magnitudes or projecting gradients to remove mutually opposing components. Second, in line with the idea that activity patterns in biological synapses are not simply frozen after learning, but are gradually reorganized through processes such as homeostatic plasticity and replay, flatness-promoting objectives reshape the geometry of the loss landscape. By forcing the network to accommodate parameter perturbations during training, these methods drive the solution toward flatter local minima, which correspond to more tolerant parameter distributions and provide greater robustness to subsequent updates. Third, gradient surrogacy reflects how biological systems can rely solely on local activity and scalar feedback to update synapses. Zeroth-order estimation methods use only forward passes and task-level feedback to construct surrogate update directions when explicit gradients are unavailable, echoing how dopaminergic signals reinforce recently active synapses tagged by eligibility traces according to behavioral outcomes.

Vision-Language Continual Learning. Besides, the diversity of input modalities, especially in vision–language models that process multi-dimensional information, has proven effective. Recent work such as DMNSP [16] highlights the evolving differences in modality variance when

adapting to task-specific features. DMNSP stabilizes this adaptation by applying selective null-space gradient projection to the image-encoder adapters and by introducing a dynamic coefficient that reduces the negative effect of non-zero residual terms caused by split layer estimation. We also integrate INFTY into the optimization of the vision adapters in combination with DMNSP, as shown in Table 1. As can be seen, INFTY supports DMNSP with better accuracy across almost all tasks.

Concept Incremental Flexible Customization. A typical example is the application of latent diffusion models (LDMs), which generate high-fidelity images by incorporating large-scale collections of image–text pairs in a latent feature space and have attracted remarkable attention in image editing, artistic creation and related applications. Building on LDMs, custom diffusion models (CDMs) apply LoRA fine-tuning to customize personalized concepts according to user needs. However, the initial training of LDMs relies on static datasets, and continual adaptation to new concepts can cause forgetting of both the pretrained generative abilities and previously memorized personalized concepts. To address this problem, CIDM [17] proposes a concept consolidation loss for training, together with an elastic weight aggregation module for inference and a context-controllable synthesis strategy to tackle concept neglect in multi-concept composition. The optimization friendly nature of INFTY allows it to cooperate easily with this framework. In Figure 6, single-concepts are incrementally introduced and generated by SD-1.5 with different balance weights in CIDM. It preserves old personalized concepts and the attributes of each learned concept, while INFTY with its flatter geometry, further enhances the quality of generation.

Future Application. ******

In summary, INFTY is a biologically inspired optimization framework for continual learning, organized into three complementary components: gradient coordination, gradient geometry, and gradient surrogacy. Gradient coordination focuses on mitigating the negative impact of dominant or conflicting gradients, thereby balancing plasticity and stability across tasks. Gradient geometry aims to guide optimization toward flatter regions of the loss landscape, reducing the susceptibility of previous knowledge to changes induced by new tasks. Gradient surrogacy studies how to perform effective optimization when backpropagation is unavailable or unreliable, by constructing gradient-free or proxy-based update rules. Taken together, these three components provide a portable toolbox for implementing continual learning from the optimization perspective, while remaining flexible enough to accommodate task-specific requirements in practical deployments. Although the current integration mainly incorporates several representative and recently proposed methods, we expect more researchers and practitioners to contribute additional modules and benefit from this unified framework. Future work can further extend INFTY by drawing on other biological or algorithmic perspectives. Overall, we believe INFTY will attract increasing attention in the continual learning community and serve as a solid foundation for continual AI.

Table 2: Illustration of accuracy and forgetting metrics in class-incremental learning, where $a_{i,j}$ denotes the accuracy on dataset D_j after learning task T_i, A_i is the average accuracy over all seen datasets up to task T_i, *Last* is the final average accuracy A_N, F_j measures forgetting on dataset D_j, and *Fgt* is the overall forgetting.

Task	D_1	D_2	$\dots$	D_{N-1}	D_N	**Metric**	
T_1	$a_{1,1}$					A_1	
T_2	$a_{2,1}$	$a_{2,2}$				A_2	
$\vdots$	$\vdots$	$\vdots$	$\ddots$			$\vdots$	***Avg***
T_{N-1}	$a_{N-1,1}$	$a_{N-1,2}$	$\dots$	$a_{N-1,N-1}$		A_{N-1}	
T_N	$a_{N,1}$	$a_{N,2}$	$\dots$	$a_{N,N-1}$	$a_{N,N}$	***Last***	
Metric	F_1	F_2	$\dots$	F_{N-1}	-	***Fgt***	

4 Methods

4.1 Context sets

Dataset protocols. All continual learning experiments are conducted on the CIFAR-100 dataset with five random seeds (1993, 1995, 1997, 1999, 2001), since different splits can lead to large differences in final accuracy. We use the standard training and test sets, which contain 500 training images and 100 test images per class. All 100 classes are evenly split into 10 tasks, i.e., 10 classes are learned in each task.

Training details. The experiments in the previous two subsections and their corresponding figures are based on a ResNet-18 backbone and use the original 32×32 images with random cropping under 4-pixel padding, random horizontal flipping, color jitter on brightness, and global normalization computed over all training images. The initial task is trained for 200 epochs with an initial learning rate of 0.1, which is decayed by a factor of 0.1 at epochs 60, 120, and 170. Subsequent incremental tasks are trained for 170 epochs with the same initial learning rate and decay factor, but with milestones at epochs 80, 120, and 150. The experiments in the third subsection are initialized from a pretrained Vision Transformer, to which we append lightweight adapters. The input images are resized to 224×224 to match the input size of the `vit-base-patch16-224` backbone, and no additional transforms are applied. All tasks are fine-tuned for 20 epochs with a constant learning rate of 0.025.

Evaluation metrics. We consider a sequence of N tasks, where at each stage i the model is trained only on the dataset of the current task with limited or no access to previous task data. Let $a_{i,j}$ denote the accuracy of the model after training on task i when it is evaluated on dataset D_j. For $j < i$, $a_{i,j}$ characterizes how much knowledge about D_j is retained after learning later tasks, which is related to forgetting, and when $i = j$, $a_{i,j}$ reflects the plasticity of the model on the current task. For each stage i, we define the average accuracy over all datasets seen so far as $A_i = \frac{1}{i}\sum_{j=1}^{i} a_{i,j}$. Based on this quantity, we define the last accuracy metric as $Last = A_N$, which measures the performance of the final model on all N datasets, and we define the average accuracy metric as $Avg = \frac{1}{N}\sum_{i=1}^{N} A_i$, which summarizes the overall learning process across all incremental stages. Larger values of both $Last$ and Avg indicate better continual learning performance. The forgetting metric provides a global measure of performance degradation on the first $N-1$ datasets. For each dataset D_j with $1 \le j \le N-1$, we define $F_j = \max(a_{j,j}, a_{j+1,j}, \ldots, a_{N-1,j}) - a_{N,j}$, which quantifies the loss of knowledge about D_j after completing all N tasks, and the overall forgetting score is given by $Fgt = \frac{1}{N-1}\sum_{j=1}^{N-1} F_j$. A smaller Fgt indicates less forgetting, and a negative Fgt implies not only the absence of forgetting but also positive forward transfer from later tasks.

Architecture structure. `MEMO` is proposed to address the unfair comparison of storage budgets between exemplar-storage methods and model-storage methods. For example, keeping a copy of a ResNet-32 backbone is equivalent to storing about 603 CIFAR-100 training instances. While model-replay based approaches can achieve better performance when the memory budget is large, under a strict storage constraint it is often more effective to store exemplars instead. This paper further examines the sensitivity of different layers by analyzing gradient norms, parameter shifts, and feature similarities. The results show that shallow layers exhibit very little variation across continual learning stages, whereas deeper layers change substantially to adapt new knowledge. Based on this observation, MEMO selectively stores deep, specialized blocks while leaving shallow, generalized blocks unconstrained. In this way, memory can be saved or reallocated to replay more exemplars, which leads to a better balance between accuracy and memory usage.

`EASE` investigates architectural expansion for pretrained models when old exemplars are unavailable, with the goal of calibrating classifiers for previously seen classes. It first extracts a prototype for each class in a co-occurrence subspace where all classes are present, and stores these prototypes as compressed summaries of the old data. Then EASE builds on the insight that semantic characteristics within the same representation subspace can be shared across distribution shifts, so the prototypes of old classes in a given subspace can be approximately reconstructed from the new distribution in that subspace. More concretely, this reconstruction relies on two implicit assumptions. First, it assumes that the semantic relationships between classes, as measured in the co-occurrence subspace, remain approximately consistent across all expanded adapters, so that the similarity matrix between old and new classes can be reused as a stable cross subspace weighting rather than being specific to a single embedding space. Second, it assumes that the prototype of an old class in a new adapter subspace lies roughly in the subspace spanned by the prototypes of related new classes, meaning that each old prototype can be represented as a similarity weighted combination of new prototypes. Under these assumptions, the method can synthesize pseudo prototypes for old classes in the

latest subspace using only new class prototypes and the previously estimated similarity structure, thereby avoiding the need to store exemplars or recompute old embeddings. Lastly, EASE applies a reweighted subspace ensemble during inference, where the logits from different adapters are combined according to their similarity to the stored prototypes of former classes.

4.2 Optimization methods

Gradient Coordination. These methods aim to reduce the negative impact on convergence caused by conflicting gradient directions or highly unbalanced gradient magnitudes. By design, these methods require at least two explicit objectives, for example one for the new task and one for the knowledge of old tasks. This requirement makes them naturally compatible with regularization-based methods, where the total loss can be decomposed into a data-fitting term and a regularization term that preserves previous knowledge. They can also be applied to the regularization used to calibrate the mixed classifier in expansion-based methods, as previously described for EASE. In contrast, most replay-based methods typically optimize a single joint objective over combined old and new samples, without an explicit decomposition into multiple losses, which makes gradient coordination less straightforward to apply.

We denote the gradients for acquiring new knowledge and retaining old knowledge as g_n and g_o, respectively. PCGrad characterizes harmful gradient conflicts by three conditions: (i) the cosine similarity is negative, that is $\langle g_n, g_o \rangle < 0$; (ii) one gradient has a dominating magnitude, for example $\|g_n\| \gg \|g_o\|$ or $\|g_o\| \gg \|g_n\|$; and (iii) the multi-objective loss exhibits high curvature along the update direction.

`PCGrad` is designed to break at least one of these three conditions so that the conflicts are no longer satisfied. To this end, it projects one gradient onto the normal plane of the other. Concretely, when a conflict between g_n and g_o is detected, the gradient g_n is updated as

$$g_n \leftarrow g_n - \frac{g_n^\top g_o}{\|g_o\|^2} g_o. \tag{1}$$

In the implementation, PCGrad traverses all task gradients and applies such pairwise projections in a random order, so that each task gradient can be adjusted with respect to the others before aggregation.

`GradVac` extends the idea of gradient coordination by explicitly exploiting the similarity structure between tasks. It empirically shows that linguistic proximity between tasks influences the cosine similarity of their gradients, and that higher gradient similarity tends to correlate positively with final model performance. Moreover, these similarities exhibit a consistent evolution trend across architectures, layers, and training steps. The core idea is based on the premise that, even when the cosine similarity ϕ_{on} between two task gradients g_o and g_n is nonnegative, it can still be smaller than the desired similarity ϕ_{on}^T implied by the language relationships between the tasks. To address this, GradVac initializes a target similarity ϕ_{on}^T for each layer and adjusts g_n in the direction of g_o by

$$g_n' \leftarrow g_n + \frac{\|g_n\| \big(\phi_{on}^T \sqrt{1-\phi_{on}^2} - \phi_{on}\sqrt{1-(\phi_{on}^T)^2}\big)}{\|g_o\|\sqrt{1-(\phi_{on}^T)^2}}\, g_o, \tag{2}$$

where ϕ_{on} is the cosine similarity between g_o and g_n, and ϕ_{on}^T is maintained as an exponential moving average target during training. This update encourages more geometrically aligned gradient directions for pairs of tasks that are closer in the language space.

`CAGrad` starts from the observation that averaging task gradients $g_i = \nabla L_i(\theta)$ to optimize the mean loss $L_0(\theta) = \frac{1}{K}\sum_i L_i(\theta)$ can severely hurt some tasks, since the average gradient $g_0 = \frac{1}{K}\sum_i g_i$ may increase the loss of the worst task. To mitigate such conflicts while still optimizing L_0, CAGrad searches for an update direction d that stays close to g_0 but improves the most disadvantaged task as much as possible. It forms a convex combination $g_w = \frac{1}{K}\sum_i w_i g_i$ with non-negative weights w_i summing to one and solves

$$w^* = \arg\min_w \; g_w^\top g_0 + \sqrt{\phi}\,\|g_w\|, \quad \phi = c^2\|g_0\|^2, \tag{3}$$

where $c \in [0, 1)$ controls how far the final update may deviate from g_0. The actual update direction is then

$$g_n \leftarrow d = g_0 + \frac{\sqrt{\phi}}{\|g_w\|}\, g_w. \tag{4}$$

In this way, CAGrad coordinates conflicting gradients around the mean objective, enjoys convergence guarantees on L_0, and empirically yields more balanced performance across tasks.

`UniGrad-FS` is a gradient projection method tailored for continual learning that explicitly investigates gradient similarity across incremental tasks. It further finds that the intensity of gradient conflicts can be alleviated on flatter loss landscapes. Concretely, it first perturbs the model parameters with a small radius and computes the perturbed gradients g_n and g_o for the new and old tasks. It then reshapes g_n and g_o in a GradVac-style manner by setting a target gradient similarity S_t^* for task t and projecting the gradients toward this target. In this way, UniGrad-FS reduces gradient conflicts during optimization and enhances the flexibility and robustness of the updates by enforcing larger inter-task gradient similarity.

Gradient Geometry. This category explicitly shapes the update direction so that the optimizer converges toward flatter regions of the loss landscape. Rather than only coordinating multiple objectives, these methods probe the local neighborhood of the current parameters and construct surrogate directions that are less sensitive to small perturbations. In the context of continual learning, such flatter solutions reduce the sensitivity of the model to subsequent parameter changes, thereby improving stability when new tasks arrive. Compared with pure regularization or replay, gradient geometry operates directly at the optimizer level and is thus orthogonal to the choice of continual learning strategy, which means it can in principle be combined with expansion-based, regularization-based, or replay-based approaches.

Firstly, we denote by $g = \nabla_\theta \mathcal{L}(\theta)$ the vanilla gradient of the training loss at the current parameters.

`SAM` introduces an adversarial perturbation of the parameters at each update. For a given step, SAM finds a perturbation ϵ^* within a small norm ball that maximizes the training loss, and then updates the model using the gradient at the perturbed point,

$$g_s = \nabla_\theta \mathcal{L}(\theta + \epsilon^*), \quad g_{SAM} = g_s \tag{5}$$

where $\epsilon^* = \rho g / \|g\|$ and ρ is the perturbation radius. This procedure encourages the optimizer to move toward parameter regions where the loss remains low under small perturbations, which corresponds to flatter minima. In continual learning, SAM reduces the local sensitivity of the solution, so that later task updates are less likely to cause large performance drops on previous tasks.

`LookSAM` constructs a stabilization branch that approximates the sharpness-aware update and adds it to the vanilla gradient, thereby replacing part of the expensive SAM computation. Concretely, given the minibatch gradient g and the SAM gradient g_s, it defines a stabilization vector

$$g_v = g_s - \|g_s\| \cos(\theta) \frac{g}{\|g\|}, \quad g_{Look} = g + \alpha \frac{\|g\|}{\|g_v\|} g_v, \tag{6}$$

where $\cos(\theta)$ is the cosine similarity between g and g_s, and α is a scaling hyperparameter. This branch is reused between SAM evaluations to provide a cheap sharpness-aware correction. LookSAM further analyzes large-batch training for vision transformers, which tend to converge to sharper local minima, and designs a layer-wise LookSAM variant for this regime. Experiments show substantial speedups over vanilla SAM while essentially preserving generalization performance.

`GSAM` starts from the observation that optimizing the worst-case loss value does not always guarantee convergence to flatter regions, and refines sharpness-aware updates by explicitly reducing the surrogate gap $\mathcal{L}(\theta + \epsilon^*) - \mathcal{L}(\theta)$ together with the perturbed loss. In particular, GSAM decomposes g into components parallel and orthogonal to g_s and corrects the SAM direction by

$$g_\parallel = \frac{\langle g, g_s \rangle}{\|g_s\|^2} g_s, \quad g_\perp = g - g_\parallel, \quad g_{GSAM} = g_s - \alpha\, g_\perp, \tag{7}$$

where α is a hyperparameter to scale $g_\perp$. The parameters are updated with g_{GSAM}, which biases optimization toward flatter minima and yields more stable solutions when new tasks are learned.

`GAM` argues that SAM's zeroth-order sharpness measure, $\mathcal{L}(\theta + \epsilon^*) - \mathcal{L}(\theta)$, does not explicitly constrain the curvature around local minima and instead regularizes a first-order flatness term that controls the maximum gradient norm in a neighborhood of the solution. It defines the first-order

flatness regularizer as

$$g_f \approx \rho \nabla \big\| \nabla_\theta \mathcal{L}(\theta^{adv}) \big\|, \quad \theta^{adv} = \theta + \rho \frac{f}{\|f\|}, \quad (8)$$

where $f = \nabla \big\| \nabla_\theta \mathcal{L}(\theta) \big\|$. The final update $g_{GAM} = g + g_f$ explicitly penalizes large gradient norms in a local neighborhood and drives the optimizer toward regions with smoother curvature, thereby enhancing robustness.

`C-Flat & C-Flat++` C-Flat first extends the application of flatness-aware concepts to explaining parameter updates in continual learning. It combines the advantages of zeroth-order sharpness in SAM and first-order flatness in GAM in a simpler way controlled by a coefficient λ, via the update $g_{C-Flat} = g_s + \lambda g_f$. Both regularization terms encourage the gradient to vary slowly within the perturbation region, which corresponds to an even flatter and smoother neighborhood. Experimental results show that C-Flat achieves superior performance when integrated into various continual learning techniques. Furthermore, C-Flat++ extends C-Flat by observing that not all C-Flat steps necessarily move toward the final flat region, and therefore interleaves part of the C-Flat updates with vanilla gradient descent steps to accelerate convergence, using a sigmoid-like proxy to measure dynamic sharpness along the optimization trajectory.

Gradient Surrogacy. This family of methods addresses scenarios where exact backpropagation is unavailable, unreliable, or partially constrained. Instead of using the true gradient, they construct surrogate directions from function evaluations or forward mode signals, and then plug these surrogates into standard optimizers. In continual learning, this perspective is useful when the backward pass must respect stability constraints, when some modules are non-differentiable, or when the feedback signal is delayed or noisy.

`ZO-SGD` is the basic zeroth-order variant of stochastic gradient descent. At each step, it samples a random perturbation direction u from a simple distribution such as a Gaussian or Rademacher distribution, evaluates the loss at $\theta+\delta u$ and $\theta-\delta u$, and uses the finite difference of these losses to approximate the directional derivative along u. Multiplying this scalar estimate by u yields a stochastic estimate of the full gradient

$$\widehat{\nabla} L(\theta; u) = \frac{L(\theta + \delta u) - L(\theta - \delta u)}{2\delta} u, \quad (9)$$

where $\delta > 0$ is a small perturbation radius. This estimator is then plugged into a standard SGD-style update with learning rate η

$$\theta_{t+1} = \theta_t - \eta \, \widehat{\nabla} L(\theta_t; u_t). \quad (10)$$

`ZO-SGD-Sign` modifies ZO-SGD by discarding the magnitude of the gradient estimate and keeping only its sign. After obtaining a noisy zeroth-order estimate $\widehat{\nabla} L(\theta; u)$ as in ZO-SGD, the parameters are updated by

$$\theta_{t+1} = \theta_t - \eta \, \mathrm{sign}\big(\widehat{\nabla} L(\theta; u)\big). \quad (11)$$

ZO-SGD-Sign can probe a larger neighborhood around the current solution, escape sharp regions, and become more robust to scale mismatch in the finite-difference estimates, but this comes at the cost of more oscillatory behavior when the surrogate gradient is too noisy.

`ZO-SGD-Conserve` stabilizes zeroth-order updates by conservatively choosing the next iterate from a small set of candidate points instead of always committing to a single estimated descent step. After computing a zeroth-order gradient estimate, it forms the candidate set by

$$C_t = \Big\{ \theta_t, \ \theta_t - \eta_t \, \widehat{\nabla} L(\theta_t; u_t), \ \theta_t + \eta_t \, \widehat{\nabla} L(\theta_t; u_t) \Big\}, \quad (12)$$

and selects the next iterate by

$$\theta_{t+1} = \arg\min_{y \in C_t} L(y). \quad (13)$$

Thus the method can either move along or against the estimated zeroth-order direction, or stay at the current point if both moves increase the loss, which mitigates overly aggressive updates caused by noisy gradient estimates.

`ZO-Adam and Others.` ZO-Adam extends zeroth-order optimization with adaptive moment estimates in the spirit of Adam. The algorithm first constructs a zeroth-order gradient estimate as in ZO-SGD, then maintains exponential moving averages of this estimate and its element-wise

square

$$m_t = \beta_1 m_{t-1} + (1-\beta_1)\hat{g}_t, \quad v_t = \beta_2 v_{t-1} + (1-\beta_2)\hat{g}_t^2, \tag{14}$$

and forms an AMSGrad style preconditioner and update

$$V_t = \mathrm{Diag}\big(\max(v_t, v_{t-1})\big), \quad \theta_{t+1} = \theta_t - \eta_t \frac{m_t}{\sqrt{V_t} + \varepsilon}. \tag{15}$$

These moments define parameter-wise learning rates that rescale the surrogate gradient before each update, which compensates for heterogeneous curvature and noise levels across coordinates. Such adaptive schemes can reduce the impact of noisy surrogate directions on sensitive parameters and slow down updates along directions that are strongly constrained by stability regularizers.

`Forward-Grad` replaces backpropagation with a forward-mode surrogate that uses Jacobian–vector products (JVPs) to estimate gradient directions. At each step, it samples a random probe vector u_t and applies forward-mode automatic differentiation to compute a JVP in that direction,

$$g_{\mathrm{FGD}}(\theta_t; u_t) = \mathrm{JVP}_{\theta_t}(u_t) = \left.\frac{\partial f(\theta)}{\partial \theta}\right|_{\theta=\theta_t} u_t. \tag{16}$$

This yields an unbiased stochastic estimate of the gradient direction using only forward-mode AD, without constructing an explicit backward graph.

Data availability

All datasets used in this study are publicly available: CIFAR-100 (https://www.cs.toronto.edu/~kriz/cifar.html), CIDM (https://github.com/JiahuaDong/CIFC), and DMNSP (https://github.com/RL-VIG/DMNSP).

Code availability

The source code for reproducing and extending all experiments in this paper is publicly available under the MIT license at https://github.com/THUDM/INFTY.

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